

EE 205: Electromagnetics and Wave propagation

Course Code	EE 205
Course Name	Electromagnetics and Wave propagation
L-T-P-C	2.5-0.5-0-0
Prerequisites	None
Intended for	Discipline core for 2nd year B. Tech Electrical Engineering, VLSI students
Mutual Exclusion	None/PH 521
Approval	52nd BoA (Nov 2023)

Course Contents

- **Vector Analysis:** Basic Mathematical operations using vectors, coordinate systems, integrals (line, surface, volume) using vector functions, vector theorems, basics of partial derivatives, Laplacian, Greens functions. (9 Lectures)
- **Electrostatics:** Electrostatic in free space, Coulomb's law, Gauss Law and its applications, Potential and work done, Conductors, Dielectric, capacitance, static fields in matter, static boundary conditions. method of images. (9 Lectures)
- **Magnetostatics:** Magnetostatics in free-space, Magnetostatics force, Vector Magnetic potential, Biot-Savart law, Ampere's law, Inductance and Magnetic materials, Boundary conditions for Magnetostatics. (9 Lectures)
- **Time Varying fields:** Introduction to time varying fields, limitation of Ampere's law, Continuity relations, Maxwell's Equations and its applications, Boundary conditions for dynamic fields, use of vector potential for dynamic fields, wave equations and propagation (in isotropic and anisotropic medium). (9 Lectures)
- **Radiation and propagation principle:** Relation between guided wave and free space wave, Concept of dipoles and monopoles, radiated fields from dipoles, Far-field and near-field, Concept of group and phase velocity. Introduction to high frequency transmission line. (9 Lectures)

Laboratory Modules

- There shall be practical demonstration for certain fundamental laws of physics and its Interpretations shall be derived in mathematical form. This shall be conducted in the form of tutorial sessions to aid theoretical concepts.

Text Books

1. Mathew N.O. Sadiku, **Principles of Electromagnetics**, 4th Edition, Oxford Press, 2007.
2. David. K Cheng, **Fields and Wave Electromagnetics**, 5th Impression, Pearson Education, 2007.

References

1. R. K. Shrivastava, **Electromagnetic waves**, McGraw-Hill Education (India) Pvt Limited, 2005.
2. David. J Griffith, **Introduction to Electrodynamics**, Cambridge University Press, Republished year: 2017.
3. G. S. N Raju, **Electromagnetic Field Theory and Transmission Lines**, Pearson Education India, 2006.
4. Walter Lewin, **Lectures on Physics**, MIT.
5. J.D. Jackson, **Classical Electrodynamics**, Wiley, 1999.